ADELEKE MARVELLOUS AYOMIDE

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PROFESSIONAL PROFILE

Final-year Remote Sensing & GIS student with strong expertise in land use land cover modelling, spatial statistics, and environmental analysis. I am an emerging spatial data scientist experienced in integrating field data acquisition with advanced GIS, GIS programming and remote sensing techniques. My research integrates machine learning and spatial regression to quantify urban-driven ecological change and support evidence-based environmental planning in rapidly growing African cities.

EDUCATION

Federal University of Technology, Akure (FUTA)
B.Tech. Remote Sensing and Geoscience Information Systems (RSG)
Expected Graduation: 2026

RESEARCH & ACADEMIC PROJECTS

URBAN EXPANSION & VEGETATION CARBON DYNAMICS – AKURE PERI-URBAN REGION (1986–2025)……..*February 2026*
Young Scholar Award Research Submission

- Performed multi-temporal Landsat classification using Random Forest.
- Quantified land-use transitions across five epochs.
- Modeled RDVI-based Aboveground Biomass (AGB) and carbon density.
- Estimated carbon loss attributable to urban expansion.
- Produced transition matrices, spatial trend maps, and statistical summaries.

URBAN HEAT ISLAND (UHI) MAPPING OF AKURE SOUTH LGA… *Nov. 2025*

- Derived Land Surface Temperature (LST) from Landsat 9 using radiance conversion, brightness temperature computation, and emissivity correction.
- Calculated NDVI from Sentinel-2 and resampled to match LST spatial resolution.
- Mapped significant high-temperature, low-vegetation clusters as UHI.

SPATIAL MODELING OF AIR POLLUTION (NO_2) CONCENTRATION IN LAGOS STATE (2024)*……October2025*

- Extracted tropospheric NO_2 concentration from Sentinel-5P TROPOMI as dependent variable, resampled and harmonized predictor datasets (SRTM DEM, ESA WorldCover, WorldPop) to a common spatial resolution.

- Applied global Ordinary Least Squares (OLS) regression to identify dominant predictors of NO_2 distribution.
- Applied Geographically Weighted Regression (GWR) using adaptive Gaussian kernel to capture spatial non-stationarity.
- Mapped spatially varying regression coefficients to reveal localized pollution drivers.

VEGETATION HEALTH & CARBON STOCK ASSESSMENT – FORESTRY RESEARCH INSTITUTE OF NIGERIA (FRIN), IBADAN (2020, 2023, 2025).. *August 2025*

- Acquired and preprocessed multi-temporal Landsat imagery covering FRIN boundary and derived NDVI of FRIN
- Applied an exponential NDVI–AGB empirical model calibrated from literature-supported tropical forest coefficients.
- Converted Aboveground Biomass (AGB) to carbon stock using IPCC-recommended carbon fraction (0.47) and generated spatially explicit AGB (t/ha) and carbon density (tC/ha) maps for 2020, 2023, and 2025.
- Conducted temporal differencing to quantify vegetation gain/loss and carbon sequestration trends.

VEGETATION CARBON STORAGE VS URBANIZATION – OSOGBO PERI-URBAN STUDY (1986–2022*)……..……..August 2025*

- Processed multi-epoch Landsat imagery (1986, 2002, 2006, 2016, 2022).
- Computed Renormalized Difference Vegetation Index (RDVI) to improve biomass sensitivity in heterogeneous peri-urban landscapes.
- Implemented exponential RDVI–AGB modeling to reduce saturation limitations common in NDVI-based biomass estimation.
- Converted biomass to carbon density and standardized outputs to tC/ha.
- Generated LGA-level carbon stock summaries using zonal statistical analysis.

GEOSTATISTICAL OZONE PREDICTION (CALIFORNIA)…… *March 2025*

- Applied Ordinary Kriging interpolation for ozone concentration modeling.
- Assessed spatial prediction accuracy and semivariogram fitting.

INDEPENDENT FIELD MAPPING & – AKURE SOUTH LGA: Course: RSG 320 – Independent Field Mapping | Group Leader……*September 2024*

- Led a structured geospatial field survey (0.35 km × 0.08 km).
- Built infrastructure databases (schools, transformers, healthcare, markets).
- Collected GPS coordinates and elevation using KoboCollect and GPS devices and measured groundwater depth in wells.
- Conducted Kernel Density, Buffer Analysis, IDW interpolation, Hotspot Analysis, and OLS regression using ArcMap and Python (Google Colab)

TECHNICAL SKILLLS

GIS & Remote Sensing
ArcGIS Pro | ArcMap 10.8 | QGIS | Google Earth Engine

Programming & Data Science
Python (GeoPandas, Rasterio, Pandas, Matplotlib) | R (Spatial modeling) | Google Colab

CERTIFICATIONS & TRAINING

- **Spatial Analysis and Satellite Imagery in GIS -** Coursera, University of Toronto, Jul 24, 2025
- **Spatial Data Science: The New Frontier in Analytics-** Esri MOOC, October 28, 2025
- **Geospatial Analysis with ArcGIS** – Coursera, University of California 2025
- **ArcGIS Pro Basics** - Esri Course, October 3, 2025
- **ArcGIS Desktop for Spatial Analysis (Basic to Pro) -** Coursera, Packt, Aug 27, 2025
- **GIS Data Acquisition and Map Design -**Coursera, University of Toronto, August 14, 2025
- **Spatial Data Science and Applications** - Esri MOOC
- **Mastering Geospatial Analysis with QGIS** - Coursera, L&T EduTech,
- **Cartography** - Esri MOOC, October 3, 2025
- **The Data Scientist's Toolbox -** Coursera, John Hopkins University, June 29, 2025

LEADERSHIP EXPERIENCE

Group Leader – Independent Field Mapping Project (RSG 320)- 2024

- Coordinated team logistics and task allocation.
- Supervised data quality control and spatial analysis workflow.
- Led technical report compilation and final presentation.

YouthMappers FUTA Chapter

- Project Manager (2024)
- Vice President (2025)
- President (2026–Present

RESEARCH INTERESTS

Spatial Data Science • Urban Growth Modeling • GIS Programming • Environmental Sustainability • Geospatial Artificial Intelligence

REFERENCES

Available upon request.